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Project Proposal — Personal Finance Manager

Goal Statement: The problem to be solved is helping busy professionals and individuals to navigate through their expenses across their cards, accounts, maximize savings, understand tax rules, and create effective budgets. Targeted Users for the AI agentic system are individuals that aren't familiar with managing their finances, middle age working parents, business owners, and freelancers with irregular income. The agentic solution would be a multi-agent system that creates custom budgets, independently gather expenses data, flag fraud, suggest how to maximize savings and evaluate spending patterns through a web interface as a conversation.

Problem Statement and Motivation: As we know many Americans live paycheck to paycheck and they do not verify if every dollar has a correct purpose like subscriptions. Mint mobile is one of the current solutions to budgeting, but it can take a lot of work like manual categorization and managing variable income. Given the problem I am wanting to solve, I wanted to confirm that an agentic AI solution was the best choice compared to human financial advisors or static apps. I chose an agentic AI system to solve finance problems because the agent would monitor accounts, renegotiate bills automatically and detect fraud all on its own. It would adapt over time and learn about the users' goals while still being scalable enough to handle complex accounts. Compared to an external and human financial manager services there is no workflow lock in or high hourly charges.

Use Case and Scenarios: A user has recently graduated from college and got their first credit card but wants to confirm if they have fraud on their card. To evaluate this for the user the agentic AI system will use an analyzer agent to scan the user's transactions to find anything unusual. It ends up flagging a \$500 Amazon purchase since the user doesn't shop with Amazon. It will cross reference the transactions to verify with confidence if it is fraud. The planner agent will create actions to resolve the issue like creating a credit charge dispute letter and suggesting credit freeze. The critic agent will review all the insight to conclude on a verified recommendation. The optimizer agent will find a list of some sign of fraud to look out for or ways to prevent it from happening. The output will be a breakdown of the fraud and recommendations like the pdf draft of the dispute letter.

Another user is a parent juggling work and taking care of their kids, they want help with creating a monthly budget to make sure their money is being spent wisely. To create this monthly budget for the user, a data fetcher agent will pull the last 3 months of bank data to provide to the analyzer agent to identify how much they are overspending in grocery shopping. The planner agent will create a \$650 grocery budget for the user and their kids. The difference in the new budget compared to last month's spending should be reallocated to a recommended fund on their budget. The critic agent is going to review all the recommendations based on the user's specific needs to conclude on the output which would be a budget pdf and email alerts for limit.

Initial System Design: The design will incorporate a 5 multi-agent workflow. Data fetcher agent will securely pull bank data and use tools like csv upload and mock plaid. Analyzer agent's role is to categorize spending and find spending patterns; to help it do that it will use the ml classifier and pandas tools. Planner agent will use optimization algorithms to create goal trackers and budgets. The critic agent's goal is to evaluate the recommendations through stress-tests by using scenario simulations. Optimizer agent uses bill negotiation and subscription APIs to find savings for the user. Other tools that I may use in my project are pdf generation for the reports, google calendar for the bill reminders, and email APIs for alerts. The system will be broken down into backend, frontend, data, and deployment.

Feasibility and Risks: The personal financial manager agentic AI system is very feasibility to create. The potential challenges that may present themselves hallucinations, data privacy, API rate limits and, legality of the advice. To help alleviate some of the challenge the system would have local-first processing to protect data, human review/critic agent review to help with the hallucinations, a disclaimer that clearly states that the system isn't financial advice and, incorporating batch processing and fallback csv to help with the API rate limits. The constraints that I may face would be privacy, scope and compute. So, there will only be 3 use cases max, local deployment and no real bank data. Timeline for the project should take about 5-6 weeks. Week 1 will focus on setting up the analyzer and data fetcher agent. Week 2 the planner agent and frontend integration will be created. Week 3 will focus on the optimizer and critic agent. Week 4 is when I'll test the scenarios and evaluate the metrics. Week 5 and 6 is the final stretch so it will consist of the demo preparation, creating the final presentation and documentation.